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In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score
```

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In [2]: df = pd.read_csv('house_price.csv')
df.head()
```

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Out[2]:
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	House Size	Bedrooms	Age of House	House Price
0	1500	3	10	350000
1	1800	3	8	400000
2	2000	4	5	450000
3	2200	4	2	475000
4	2500	5	15	525000

```
In [3]: X = df[['House Size', 'Bedrooms', 'Age of House']]
Y = df['House Price']
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In [4]: X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_
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In [5]: model = LinearRegression()
model.fit(X_train, Y_train)
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Out[5]: + LinearRegression
        ► Parameters
```

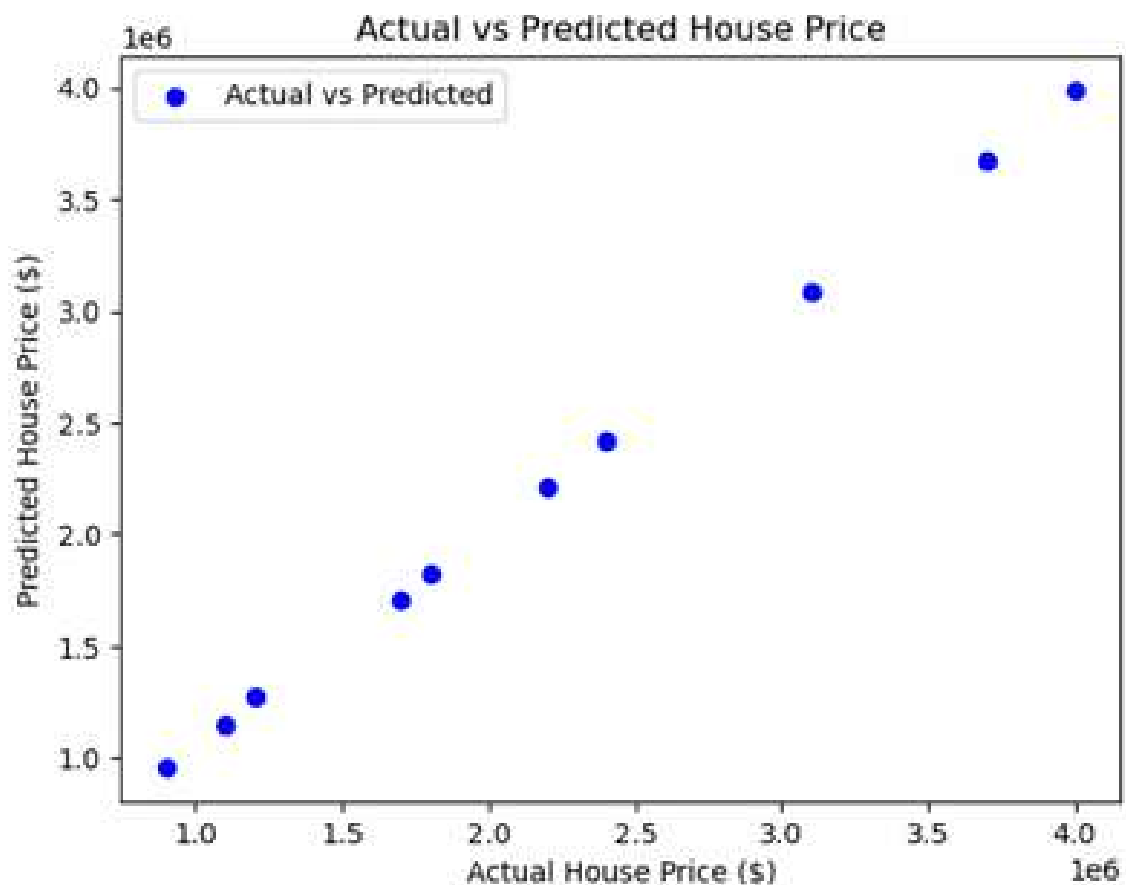
```
In [6]: Y_pred = model.predict(X_test)
Y_pred
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Out[6]: array([ 958198.91326438, 3884757.95178289, 2213923.8591368 ,
        3676539.75229267, 1145827.56632132, 3989268.46866852,
        1827657.81346492, 1786685.73548826, 2419439.6139755 ,
        1275855.59897449])
```

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In [7]: r2_score_value = r2_score(Y_test, Y_pred)
r2_score_value
print(f"R-squared Score: {r2_score_value}")
```

R-squared Score: 0.9987416757128748

```
In [8]: plt.scatter(Y_test, Y_pred, color='blue', label='Actual vs Predicted')
plt.xlabel('Actual House Price ($)')
plt.ylabel('Predicted House Price ($)')
plt.title('Actual vs Predicted House Price')
plt.legend()
plt.show()
```



```
In [9]: user_House_Size = int(input("Enter House Size: "))
user_Bedrooms = int(input("Enter Number of Bedrooms: "))
user_Age = int(input("Enter Age of House: "))
user_df = pd.DataFrame([[user_House_Size, user_Bedrooms, user_Age]], columns=["H
predicted_price = model.predict(user_df)
print(f"The predicted House Price is: ${predicted_price[0]:.2f}")
```

The predicted House Price is: \$62901847.80

In []: